

AI-POWERED VIDEO EDIT PRO

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Abstract—

Video editing has always been a time-consuming task. With increase in people that want to create and share content worldwide, video editing's complexity has been well recognized by now. People that are new, or with lack of experience mostly fail to align with modern interfaces of video editing software to carry out their required task. Our paper introduces AI- powered Video Edit Pro, an AI-based model with simple interface and organized workflow. Our model portrays optimized automation of repetitive yet necessary tasks in process of video editing along with incorporated intelligence through deep learning and Machine Learning algorithms. Convolutional and recurrent neural networks are used in different ways to implement different features of automation process. Each individual algorithm of a particular feature is triggered by either direct user selection or through analyzing user's prompt. Automated features for example are silence removal, noise reduction, color correction, overlay of text, collage of videos, audio and video enhancement etc. With help of automation through intelligence and tailored flow of actions, people who seek to make their content more professional will receive huge assistance. Findings along the development process enabled us to see potential usage of AI technologies that are going to revolutionize video editing with high level advancements in future.

Keywords— Video Editing, AI Automation, Machine Learning, Scene Detection, Real-Time Processing.

I. INTRODUCTION

Video editing plays an essential role in the digital content creation industry, yet it remains a challenging and time-consuming task. Prompt based video editing can save users time and efforts by a big difference. The user's prompts are analyzed by light weight natural language processing (NLP) which is implemented via recurrent neural network, regular expressions, fuzzy matching for the prompts. Our model provides users to [1-5]

customize the interface upon their choice so that they can regulate their ease of editing with AI-Powered video Edit Pro. Features are suggested to user with regard to his prior usage with help of predictive methods that showcase correlation with the assistance he needs. For video manipulation, the core python library we used is moviepy, along with open cv. Audio manipulation on the other hand is handled with pygame, pedalboard, librosa, pydub and wave libraries of python. The initial user interface consists of a video player container (for preview), available (unsorted) features(unsorted) as buttons. And a text field that takes user prompts then parses it with lightweight NLP algorithms mentioned earlier. Reducing or restricting manual intervention to minimal amounts. This saves users time and energy and allows user to focus more on video creation rather than stressing. Adaptability of model to modern tools and user demands is very promising. Content creators can access these customizable features with just few clicks and achieve results without giving much attention to detail. Which in turn return many more possibilities for creators to express themselves.

II. LITERATURE SURVEY

The papers reviewed for our project are mainly focused on automation techniques to enhance and apply video processing features needed by user.

M.Abovalola (2024) his study explored the realm of automation of editing using deep learning algorithms. They were able to assign neural network to do complex task. Realtime editing and high computational requirements were challenges of his approach.

J. Doe (2024) used Generative adversarial networks (GAN'S) to achieve creative counters to enhance video quality. They reported that video quality can significantly uplifted with help of GANS. High level computational and large datasets doesn't make their study optimal for everyday users.

A. Smith (2023) used natural language processing(nlp) to achieve automated video summarization. The idea was to extract key theme and generate apt summaries of video content, to make them more discoverable, [6-10]

III. PROPOSED SYSTEM

The functioning of the video editing process is done via a modular approach that is governed by a neural network. Each module is a package that is hard coded to accomplish a task by applying it on user provided video input. These packages are done using open-source python libraries that specialize in handling video and audio data.

Stage	Component	Description
Input Stage	User Input	Upload Video, Select Features, Provide Preferences
	Data Sources	External Data (Trends, User Preferences)
Processing Stage	AI Analysis	Scene Detection, Content Recognition, Editing Suggestions
	Editing Operations	Automated Editing, Effects Application, Audio Adjustments
Output Stage	Final Video Compilation	Rendering Video, Export Options
	User Feedback	User Reviews, Improvement Suggestions
Feedback Mechanism	User Interaction	Update Preferences, Submit Feedback
	Continuous Learning	Improve Algorithms, Adapt AI Models

Fig 1. Data Flow Representation

In the initial stages of the process the user selects video/videos of his choice ,the video input is then checked if it needs to be treated with preprocessing steps for productive editing .Features in this model have their own individual coded packages that work with others in order to get a specific edit. The initial testing suggests few transformations to be applied ,the rest are done by understanding user prompts.[11-15]

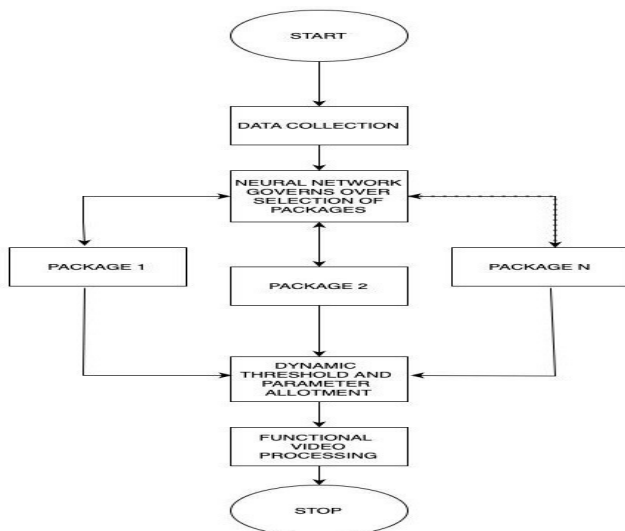


Fig 2. Data Flow Representation

During the next phase we utilize the diverse available predefined resource codes for features ,constructively clashing with user prompt of tastes ,to know the needs of user accurately .expertly devised algorithms and highly

functional apt codes are used to achieve results that align with user request. The user request is then processed with help of input analysis by regular expressions and machine learning algorithms. They score the input to judge what feature needs to be applied.

In the final stages ,post editing operations like adding music, stabilizing video, optimizing audio levels and allowing users to manipulate video resolution based on their requirement is done .Templates that are commonly used like subscribe for YouTube or follow for Instagram and also title cards are added if specified by user.

Overall process monitoring is done by a global neural network that is installed to select apt procedures and return process stats and efficiency for evaluation of proposed methodology and developed model.[15-19]

IV. RESULT

This section depicts the functional efficiency & adaptability of Video Editor Pro system. The parameters developed in the study are used for evaluation accuracy, user feedback, process speed, and model portability.

A. System Preferences Performance

The model has shown an accurate feature suggestion rate of over 90%, which shows neural networks' decisiveness to be precise, concise in terms of choosing a package to apply and conforming to user needs. Prompts of the user has been aligned with artificial intelligence's decision and the right selections have been made.

B. Evaluation of Results

The stats indicate this model factors in easing user's schedule for minimizing user effort in creating videos of great quality & keeps user up-to-speed in market trends. Optimal integration of features & techniques. The video editor is well-suited for vlogs & content creation, well-suited for social media. Freeing users of repetitive redundant tasks that take up a lot of time and effort.

V. CONCLUSION

In conclusion, our project makes sure that the job of content creators is kept to creation and not processing and post-production. Editing leads for the user to be more productive and creative in their respective field. This is also accompanied by reducing the costs and amount spent on editing. The study also acknowledges that we found limitations in the project, for what extents our model is applicable to modern editing scenes. Complexity is identified, but it still is a tedious task to apply deep learning on video editing, and time taken to process is not consistent with all systems.

VI. FUTURE WORK

Looking ahead, In field of video editing's automation through Artificial Intelligence there are many ways significant aspects to further enhance the process by AI powered tools and advanced video and audio manipulation techniques. Some regions of development in tis context are AI generated content(thumbnails, stickers ,overlays), remote editing achieved by cloud storage and processing, Realtime collaborative editing with high fault that can withhold many editors at once. Although we have seen surge in approaches to handle video editing tasks the key problems like

allotment of resources, time of computation, user satisfiability and modal accuracy have fell short than performance expected. Integrating various tools to manipulate video and audio has been very challenging in the past ,but as more research on this particular agenda is done because of rise in numbers of content creators, we might have simpler and faster data processing techniques. Another exiting aspect is usage of CUDA(Compute Unified Device Architecture) a parallel computing platform developed by NVIDIA allows developers to use computing. With help of CUDA computing we can expect fast, parallel processing in workflow of video editing with artificial Intelligence graphic processing and rendering will take significantly much less time if it is CUDA optimized .so, with help of CUDA in future results can be achieved with high throughput if paired with video editing and machine learning technologies. Finally, Engaging with the user through feedback, reviews helps in understanding the trends of market and modal performance better. The requirements of content creators are everchanging, to put up with their demands regular updates and tools of high performance must be provided. By embodying the creators vision and developers tools AI powered video editing can generate high quality content in the future.

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